

JASPER COUNTY AP, TX, USA

WMO: 720296

Lat: 30.886N

Lon: 94.035W

Elev: 213

StdP: 14.58

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53945

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.2	31.2	17.3	13.3	40.5	21.9	16.5	43.2	18.5	54.9	16.8	54.9	1.8	10	0.343

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.1	97.5	76.0	95.2	76.3	93.2	76.7	80.5	88.3	80.0	88.0	79.2	87.4	5.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.9	151.2	83.7	78.0	146.5	83.2	77.2	142.8	82.8	44.3	88.6	43.5	87.9	42.8	87.4	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.2	13.0	11.5	DB	22.8	99.4	3.5	3.2	20.3	101.7	18.3	103.5	16.4	105.3	13.8	107.6
			WB	22.1	82.4	3.6	1.8	19.6	83.7	17.5	84.7	15.5	85.7	12.9	87.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.8	50.5	54.7	61.6	67.1	74.4	80.3	81.7	83.0	78.5	69.4	59.3	52.9	(6)
	DBStd	13.55	9.58	10.14	9.00	7.07	5.96	3.62	3.20	3.56	4.54	7.79	9.27	9.72	(7)
	HDD50	290	112	60	15	2	0	0	0	0	0	0	22	79	(8)
	HDD65	1674	460	313	173	58	8	0	0	0	0	47	220	395	(9)
	CDD50	6804	127	191	375	514	756	910	984	1023	855	600	300	169	(10)
	CDD65	2715	10	24	68	121	299	460	519	558	405	182	48	21	(11)
	CDH74	27079	26	104	404	1008	2846	4869	5622	6305	3869	1671	291	64	(12)
	CDH80	11904	0	11	52	202	1064	2303	2693	3225	1759	550	44	1	(13)
(14) Wind	WSAvg	3.9	4.6	5.1	5.0	5.0	4.1	3.0	2.7	2.7	2.8	3.4	3.9	4.4	(14)
(15) Precipitation	PrecAvg	54.8	5.1	4.1	4.2	4.3	4.0	4.9	3.9	4.0	4.3	5.0	5.2	5.5	(15)
	PrecMax	77.5	10.9	10.7	8.5	10.1	9.7	13.0	7.8	7.9	11.5	21.1	13.3	10.9	(16)
	PrecMin	32.6	1.0	0.5	0.8	0.5	0.0	1.3	0.9	0.0	1.6	0.0	0.5	1.7	(17)
	PrecStd	10.7	3.0	2.8	2.2	2.4	2.5	2.8	1.9	2.2	2.6	5.0	3.4	2.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.0	81.8	84.1	87.8	93.2	98.9	98.3	101.5	96.5	91.4	84.9	79.5	(19)
		MCWB	66.3	68.9	68.1	70.3	73.7	74.5	75.7	75.3	75.2	73.7	72.5	70.5	(20)
	2%	DB	73.7	77.3	81.5	84.6	90.7	95.5	95.7	98.6	94.5	88.5	80.7	75.5	(21)
		MCWB	64.7	66.7	67.4	69.1	73.9	75.6	76.4	76.2	75.2	73.4	70.3	69.2	(22)
	5%	DB	71.1	73.7	78.9	82.2	88.5	92.8	93.5	95.7	91.9	86.1	77.2	73.0	(23)
		MCWB	64.7	66.5	66.9	68.5	74.3	76.8	77.4	77.1	75.2	72.7	68.1	68.4	(24)
	10%	DB	67.8	71.1	75.5	80.1	86.3	90.6	91.5	93.1	89.6	83.0	74.4	70.2	(25)
		MCWB	62.6	65.2	65.3	68.1	73.3	76.8	77.7	77.2	74.9	71.5	67.5	66.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.0	71.4	72.1	75.8	79.1	81.1	81.4	82.1	80.4	79.8	75.2	72.4	(27)
		MCDB	73.2	76.3	78.2	81.0	86.4	88.9	88.7	89.1	86.6	85.9	81.4	76.1	(28)
	2%	WB	68.5	70.2	70.9	73.8	77.4	80.0	80.3	80.7	79.2	77.8	73.4	71.0	(29)
		MCDB	71.7	74.7	76.6	79.9	85.3	88.0	88.2	88.7	87.0	83.9	77.7	74.3	(30)
	5%	WB	66.4	68.6	69.8	72.6	76.2	79.1	79.6	79.9	78.2	76.2	71.1	69.7	(31)
		MCDB	69.8	72.6	75.0	78.7	84.1	87.0	87.9	88.3	86.3	82.1	75.6	72.9	(32)
	10%	WB	63.7	66.1	68.6	71.2	75.2	78.2	78.9	79.1	77.3	74.5	68.7	66.8	(33)
		MCDB	66.4	70.2	73.8	77.0	82.8	86.0	87.1	87.6	85.3	80.0	73.1	69.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.3	20.3	21.4	22.0	20.6	19.4	19.1	20.1	20.7	22.8	22.0	19.5	(35)
		MCDBR	23.1	21.0	21.0	22.3	22.0	22.7	21.7	23.4	23.1	23.0	22.5	20.4	(36)
	5% WB	MCWBR	15.0	12.2	10.0	9.9	8.1	7.5	7.1	6.4	8.0	10.2	12.9	14.0	(37)
		MCWBR	18.1	17.4	17.4	17.6	18.9	19.1	19.5	19.6	19.7	18.3	18.5	17.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.329	0.349	0.369	0.417	0.456	0.488	0.492	0.483	0.445	0.374	0.364	0.341	(40)
		taud	2.533	2.484	2.423	2.295	2.223	2.171	2.200	2.225	2.296	2.474	2.451	2.503	(41)
	Ebn at Noon		286	290	290	279	267	257	255	257	263	277	270	275	(42)
		Edh at Noon	27	31	36	42	45	48	46	44	40	31	30	27	(43)
(44) All-Sky Solar Radiation	RadAvg		888	990	1384	1684	1865	1924	1912	1813	1580	1341	1002	772	(44)
	RadStd		82	170	122	111	135	146	150	155	147	182	104	91	(45)

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.58	N/A	N/A	N/A	N/A	N/A	N/A	+189	+149

Nomenclature: See separate page